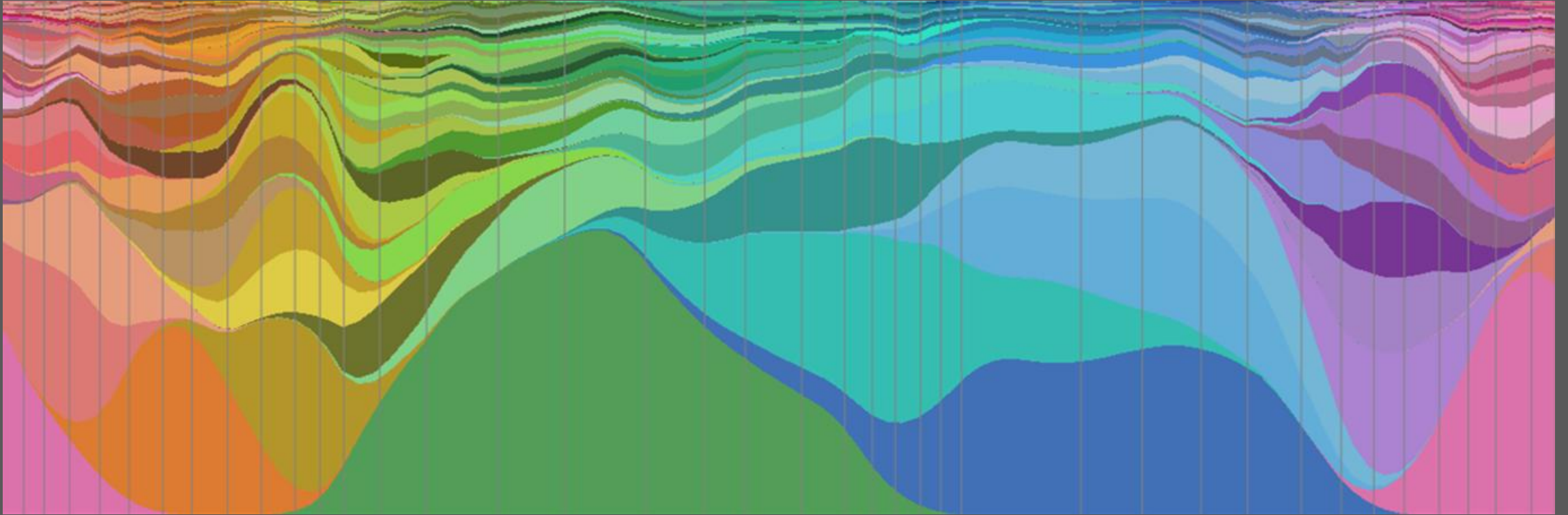


LDSCI5209 - Information Presentation and Visualisation

Week 9: Visualisations for Machine Learning



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Recap: Networks & Trees

- Networks (graphs) consist of nodes and edges that model relationships in data
- Trees are hierarchical networks with parent-child structure
- Applications include social networks, epidemiology, supply chains, decision trees, and search engines
- Visualisation challenges: overcrowding, depth vs. breadth tradeoffs, and need for interactivity in complex structures

Plan for today

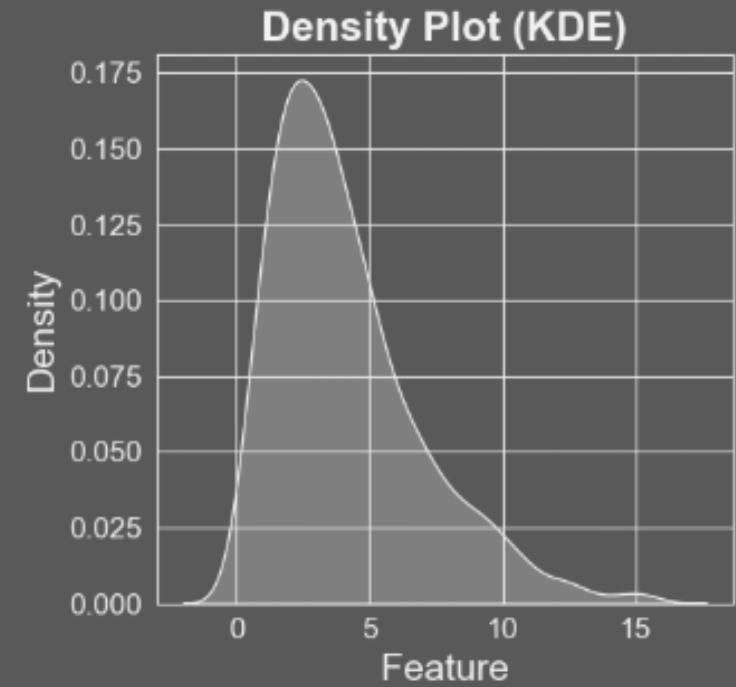
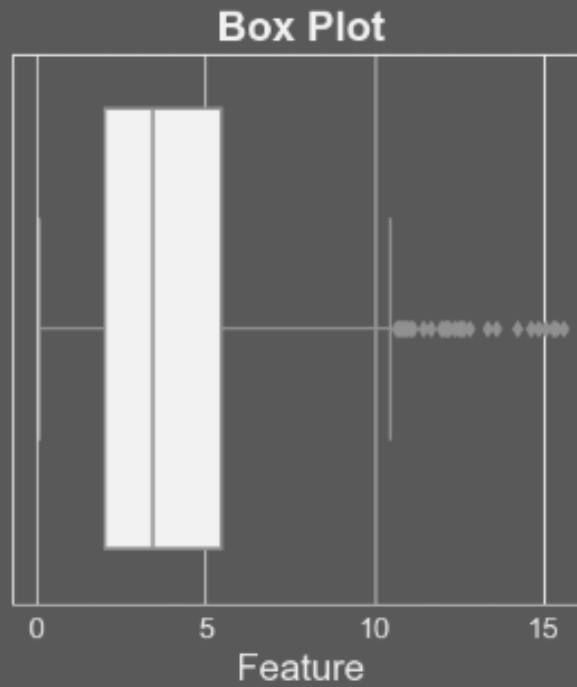
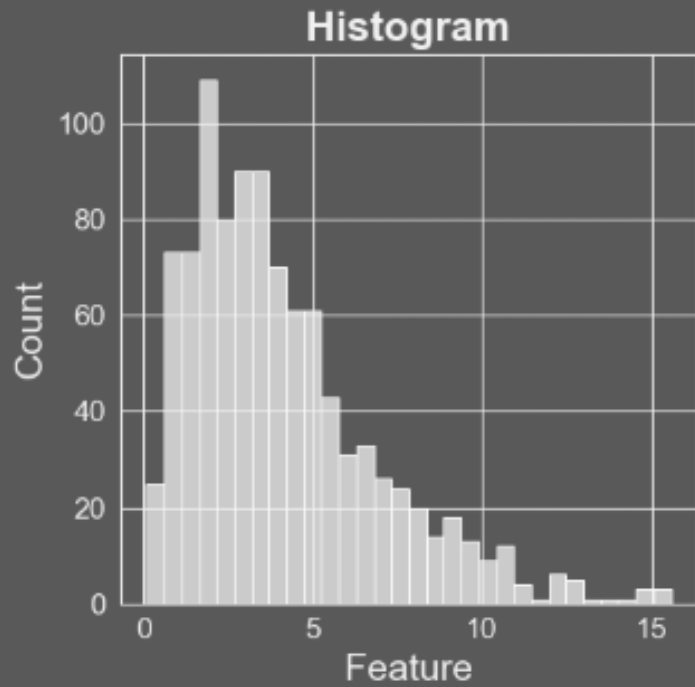
Visualisation for machine learning

- Data exploration and preprocessing
- High-dimensional data
- Feature engineering
- Model performance and hyperparameter tuning

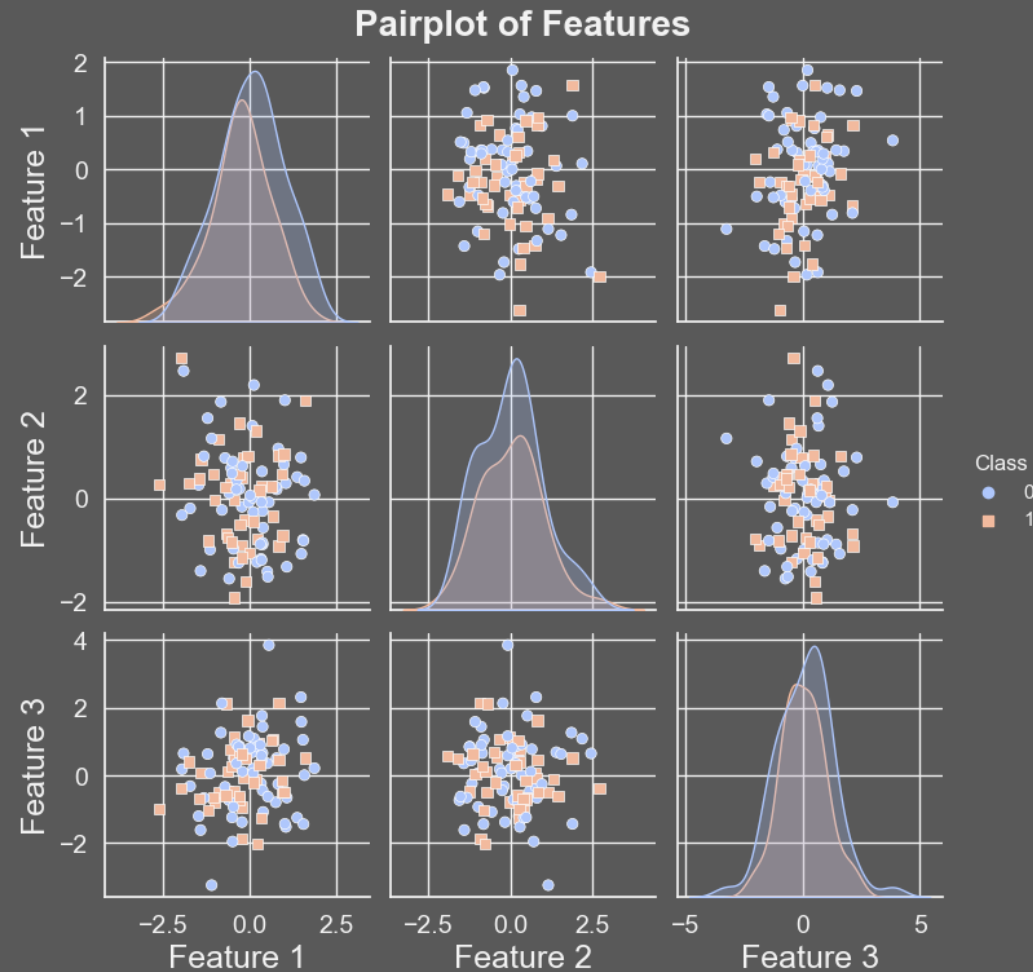
Data exploration and preprocessing

Distributions, relationships, missing data and outliers

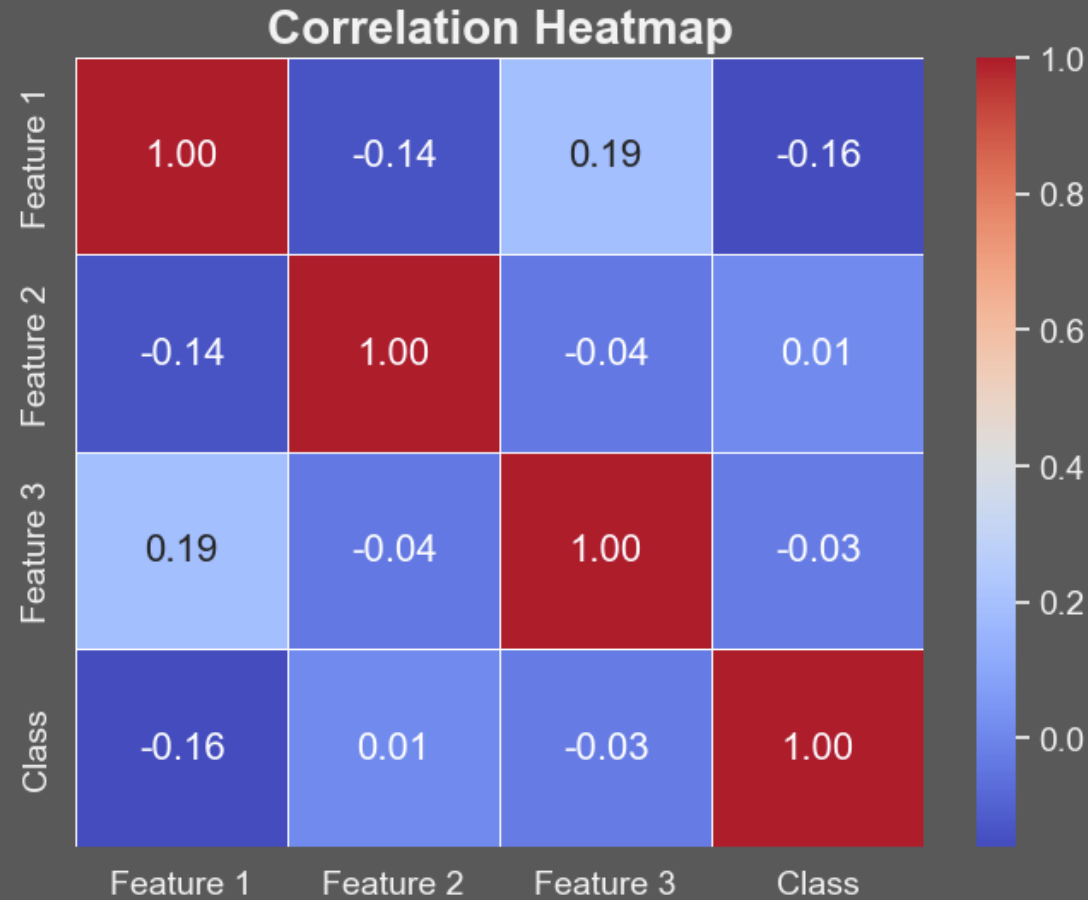
Visualising data distributions



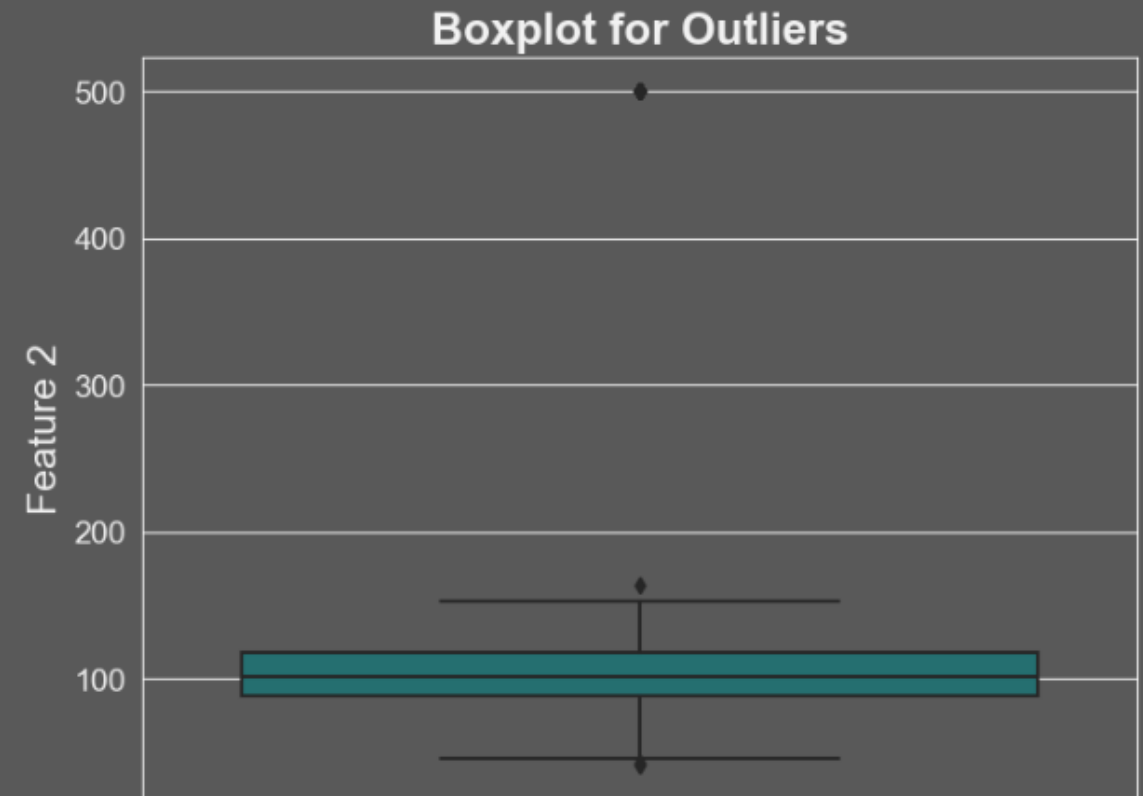
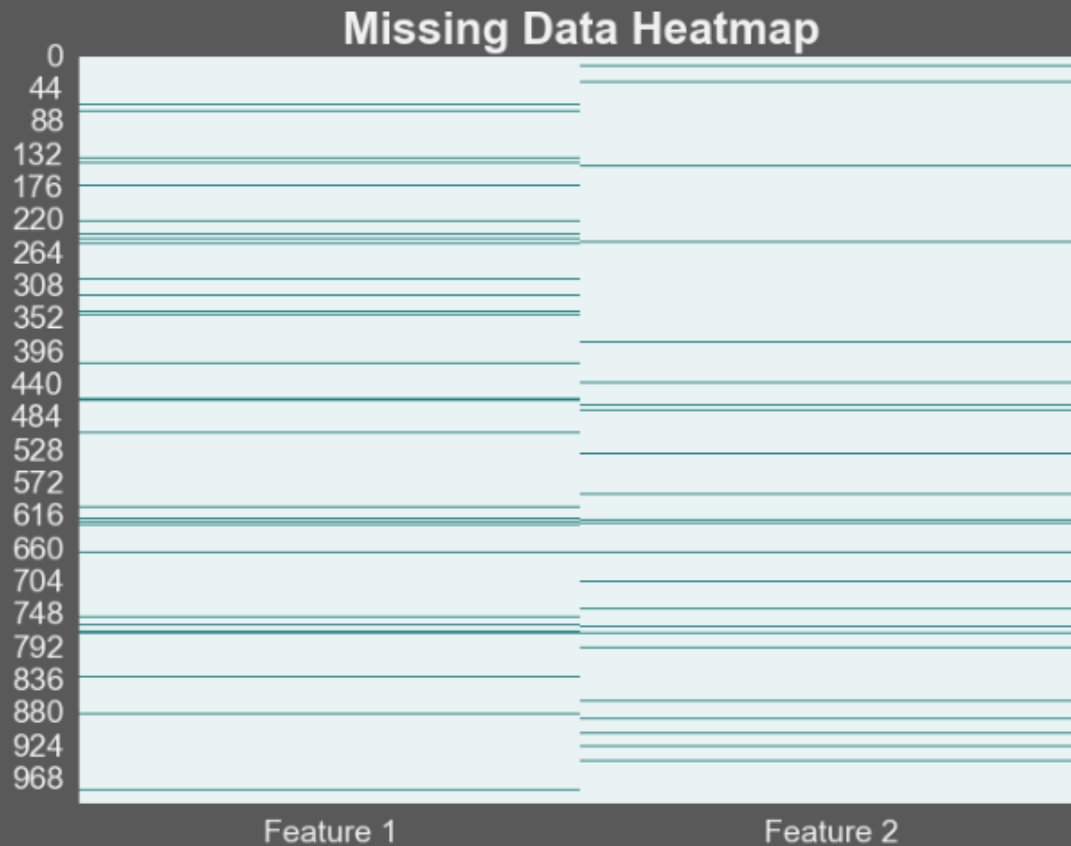
Relationship between features



Relationship between features



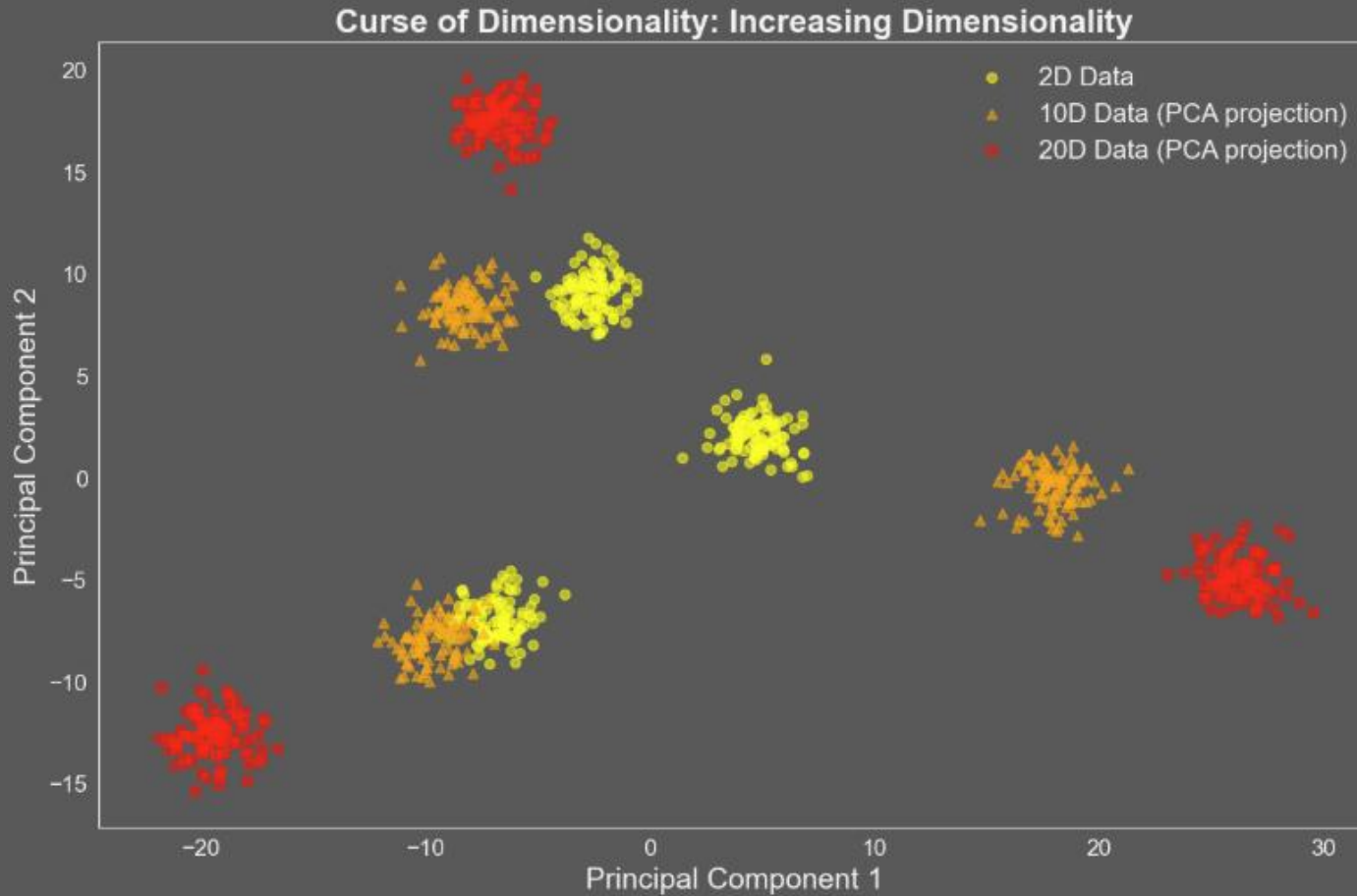
Handling missing data and outliers



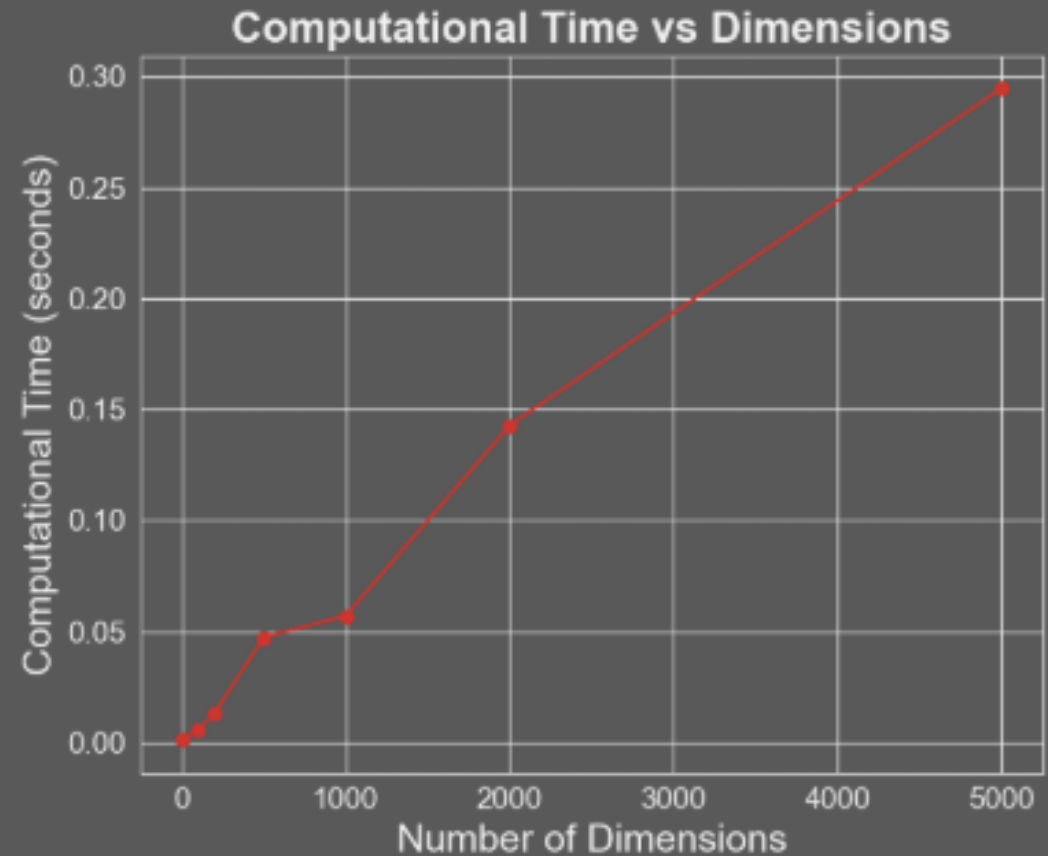
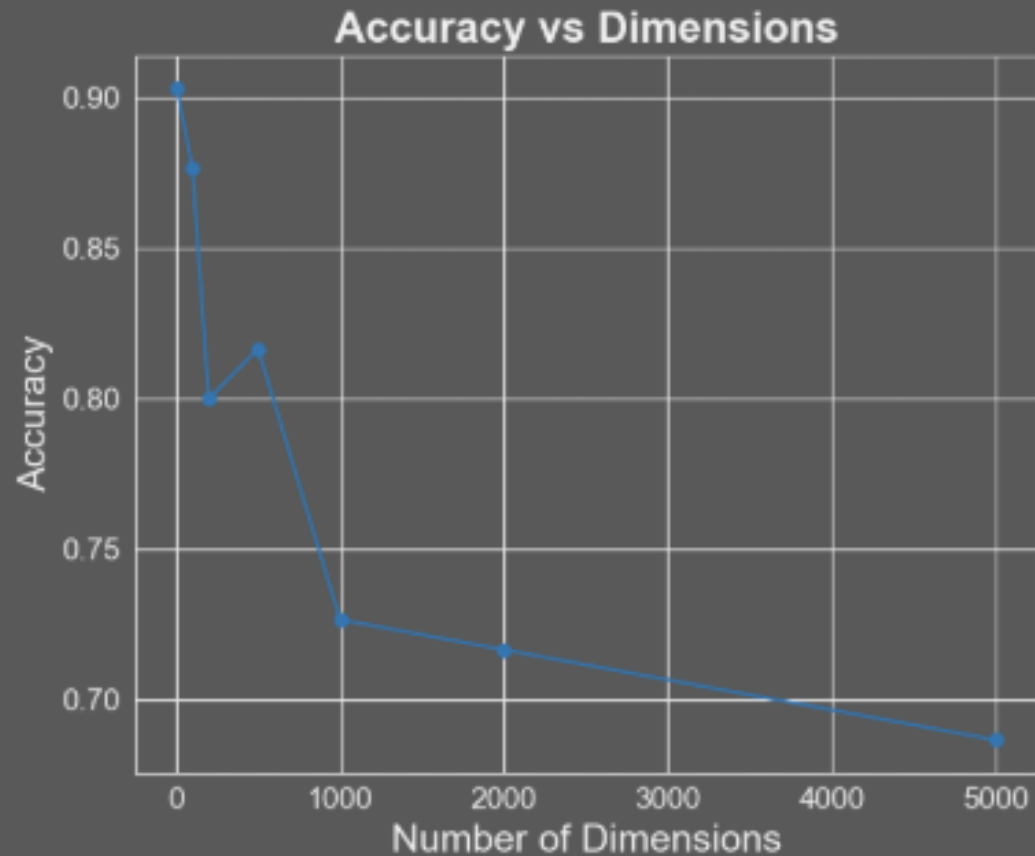
High-dimensional data

Curse of dimensionality, reduction techniques, embeddings

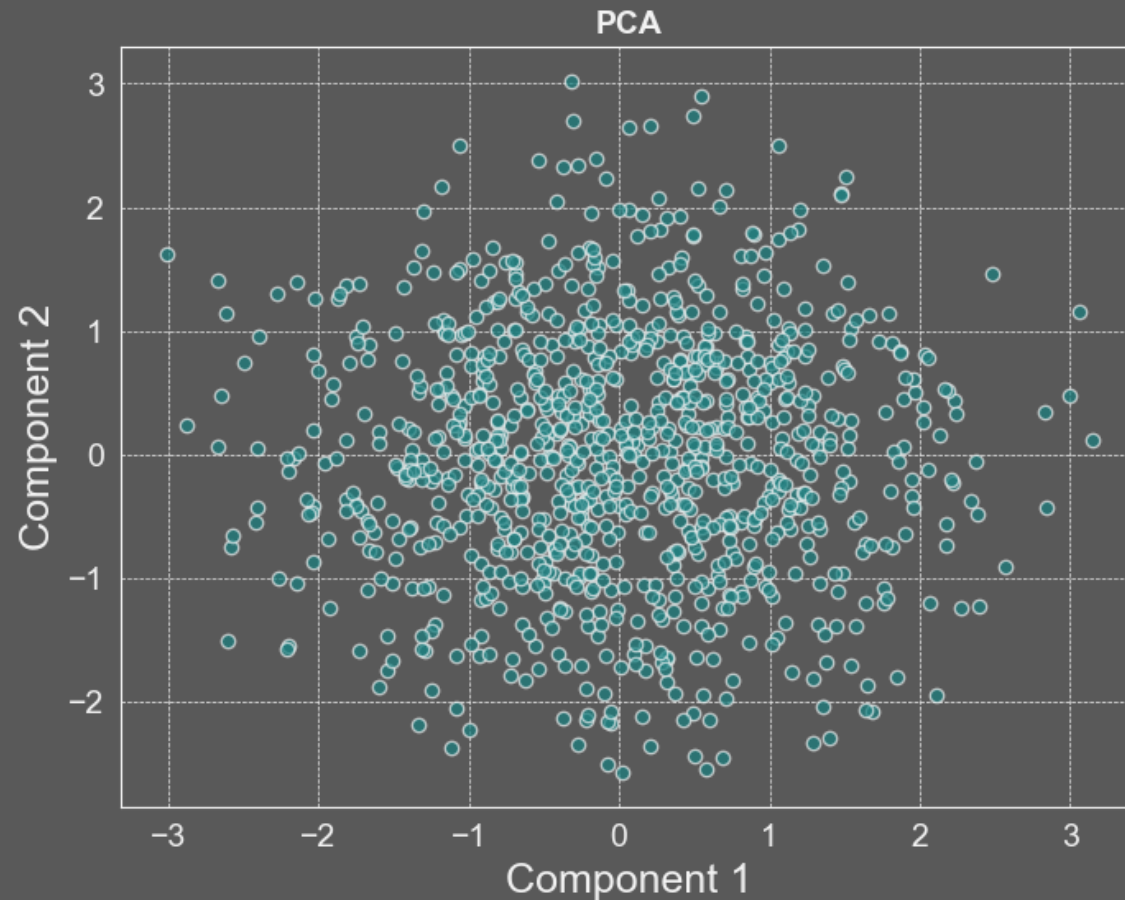
Interpretability cost



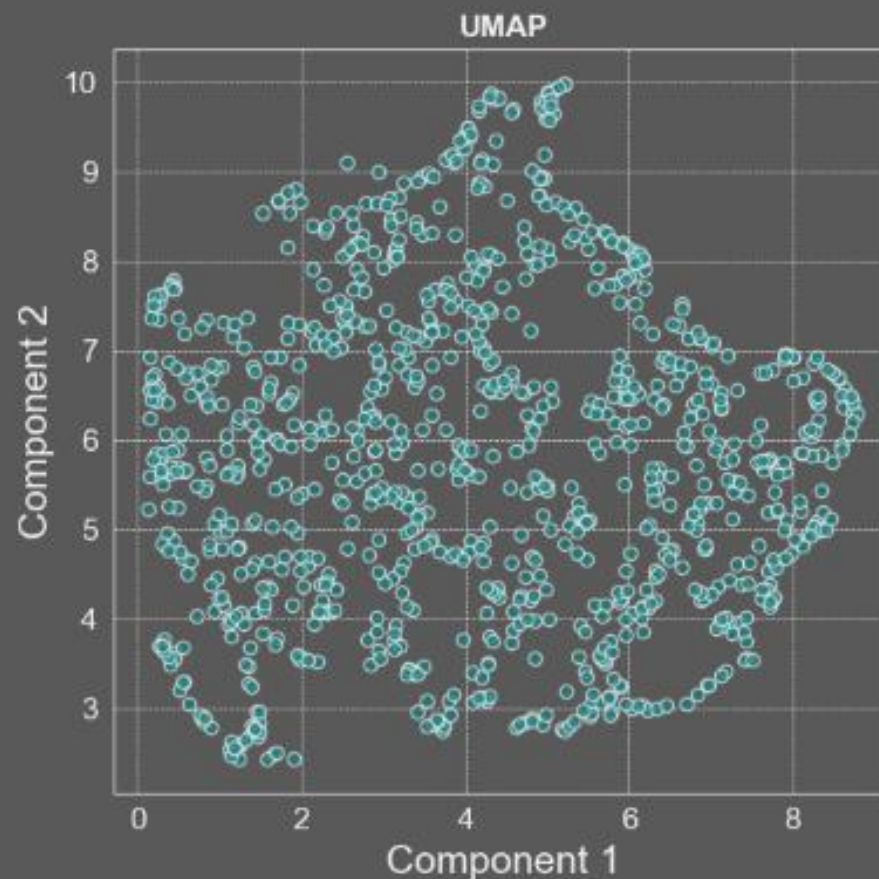
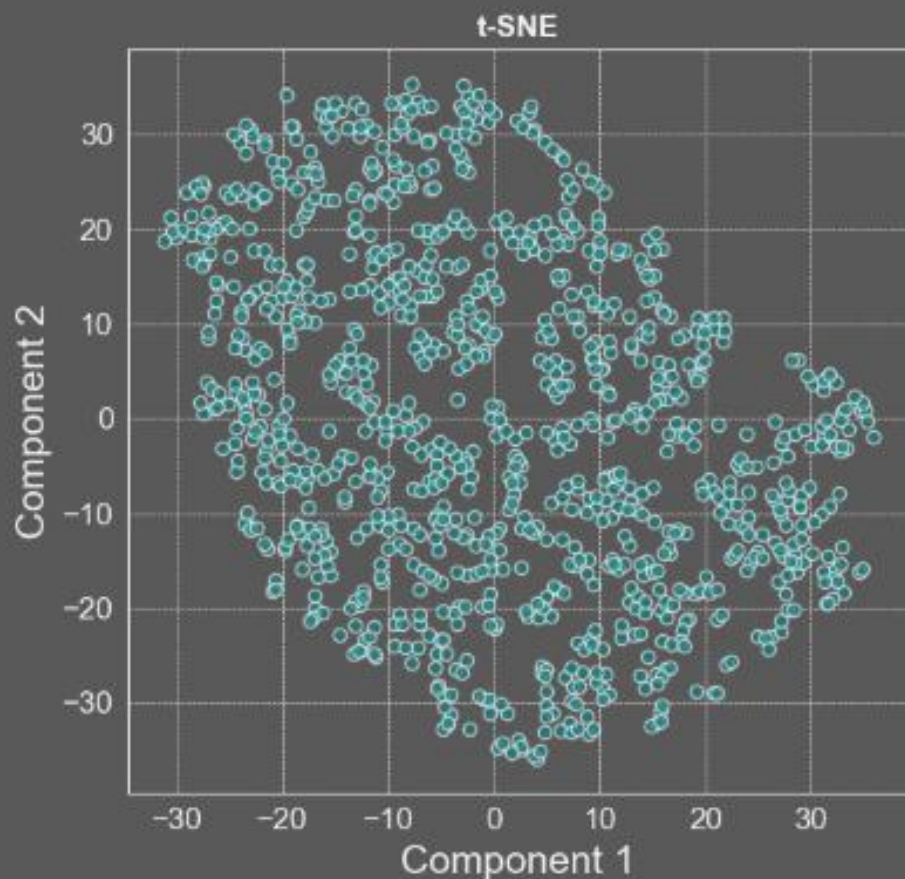
Performance and computational cost



Dimensionality reduction techniques



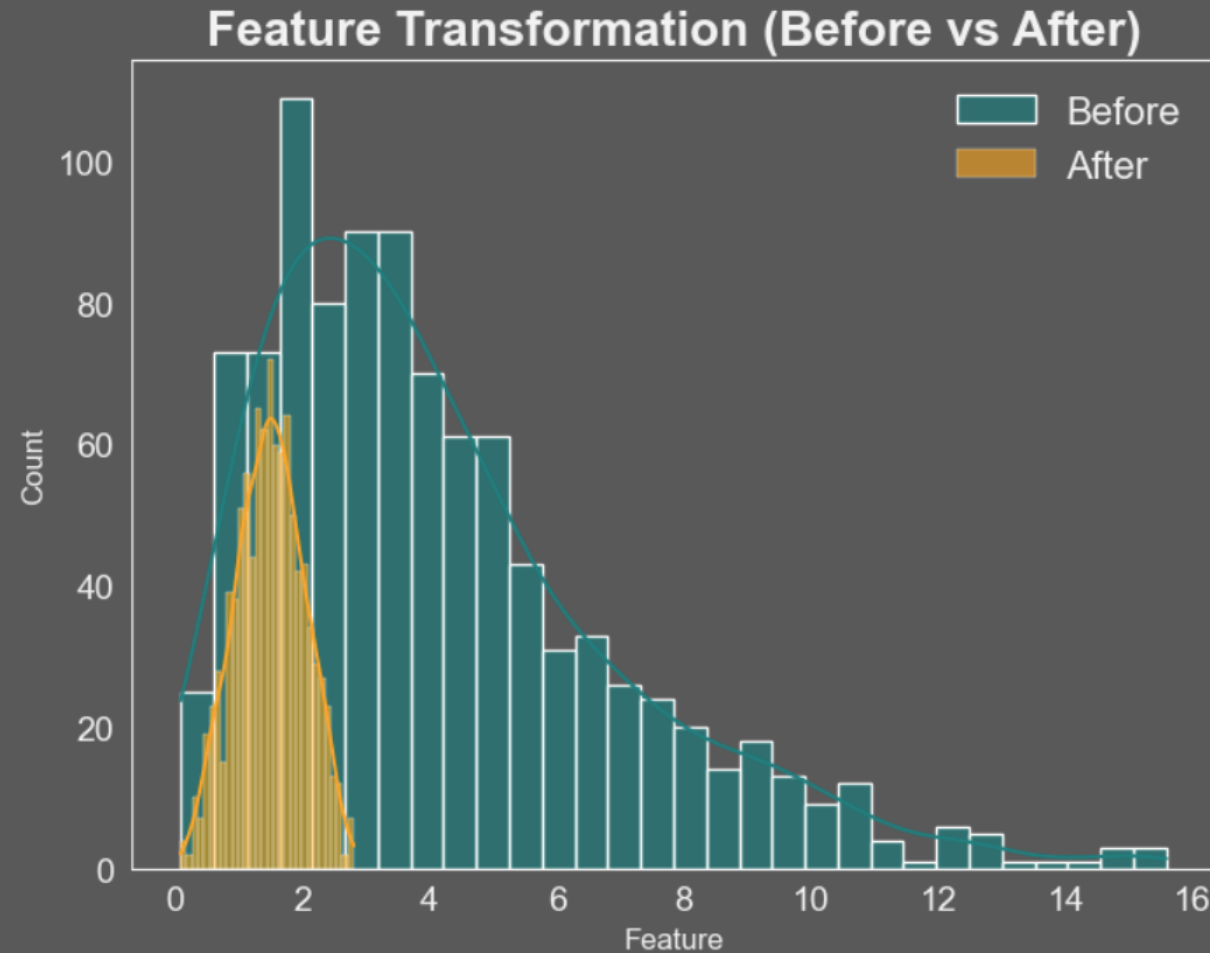
Dimensionality reduction techniques



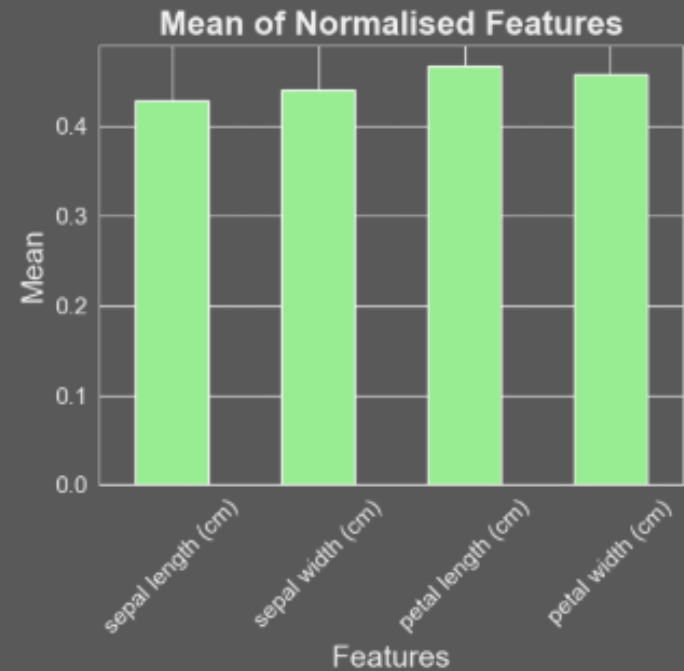
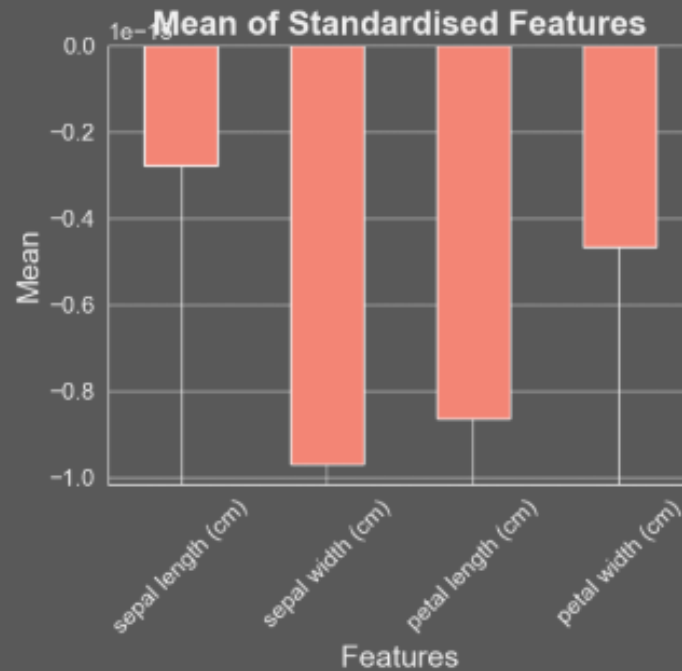
Feature engineering

Distributions, transformations, importance, selection

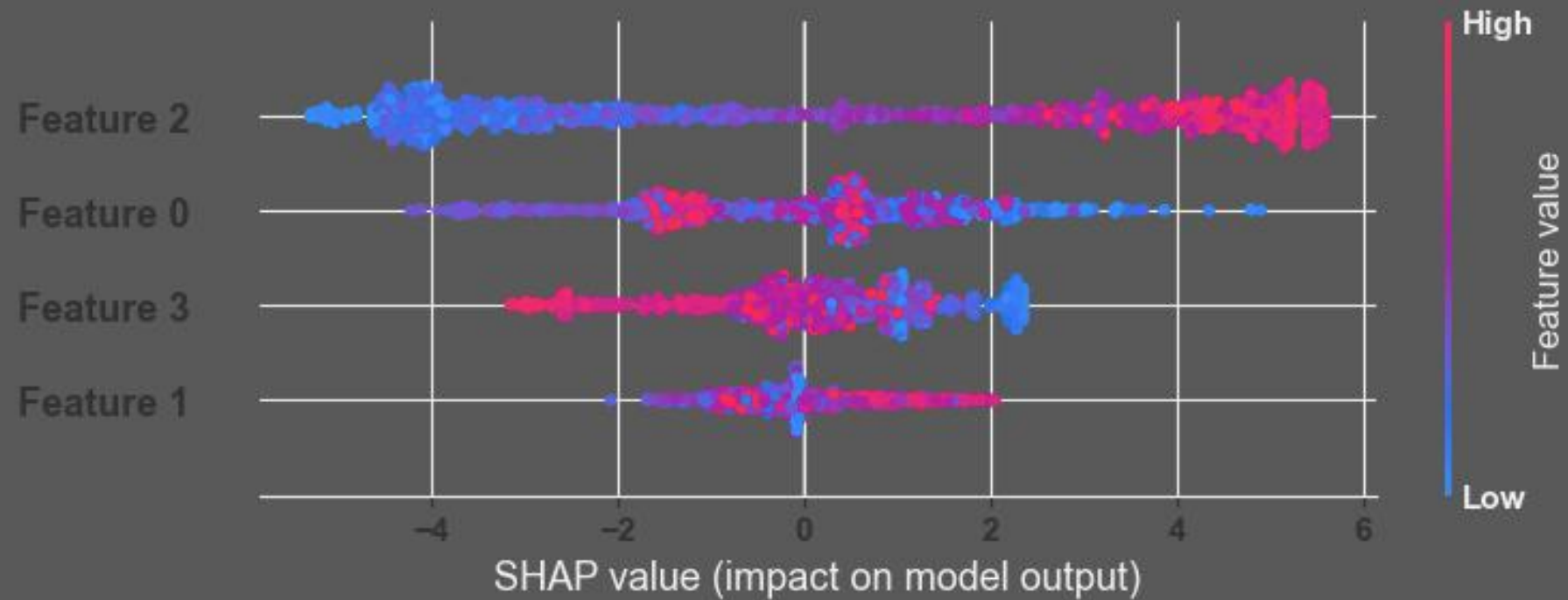
Feature distribution and transformations



Feature transformations



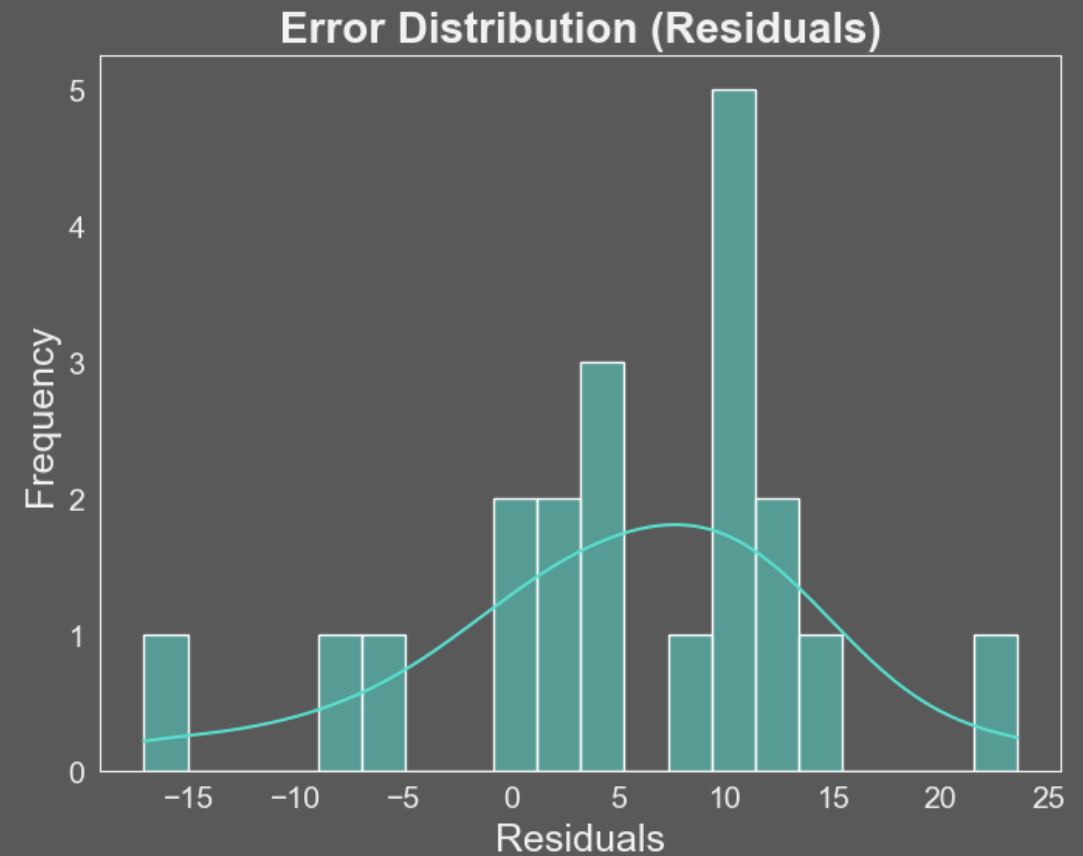
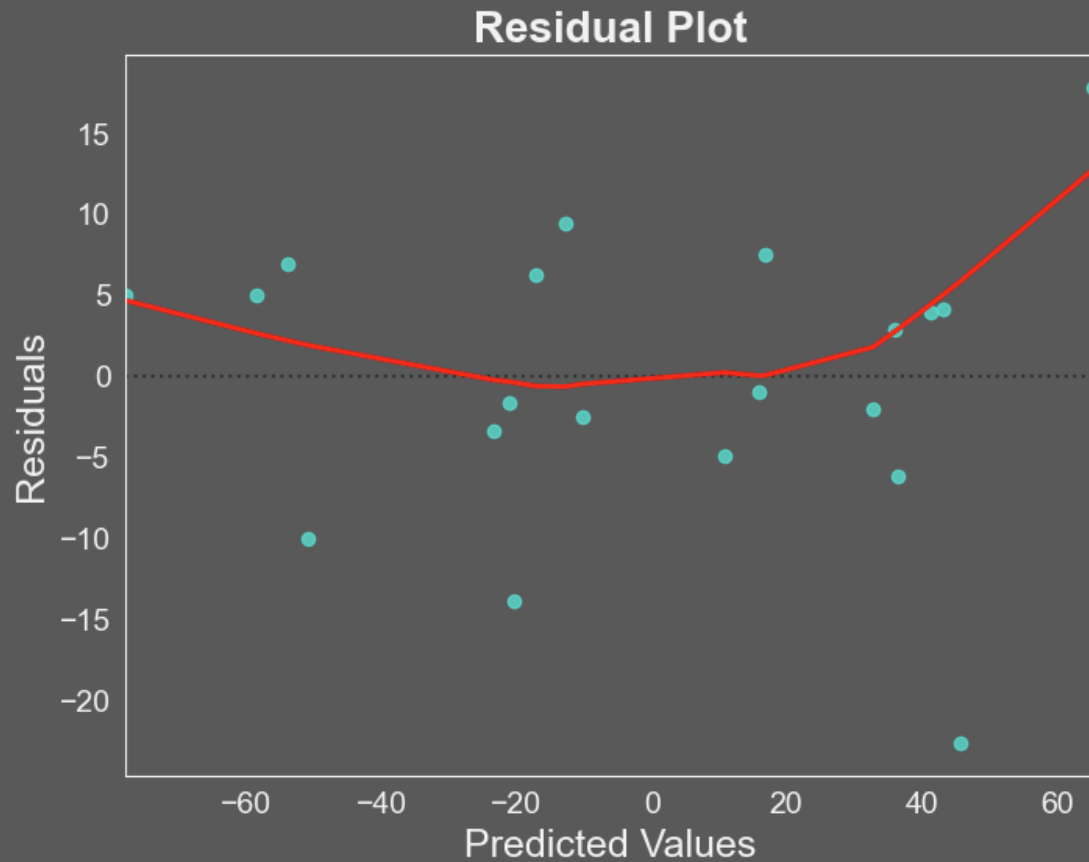
Feature selection: Feature importance



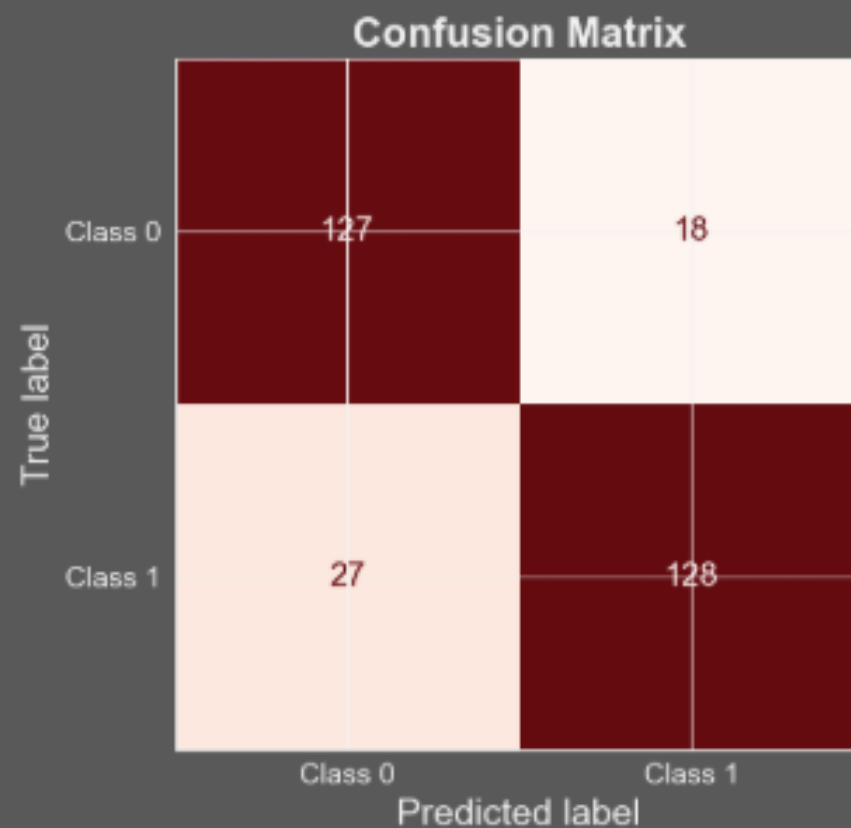
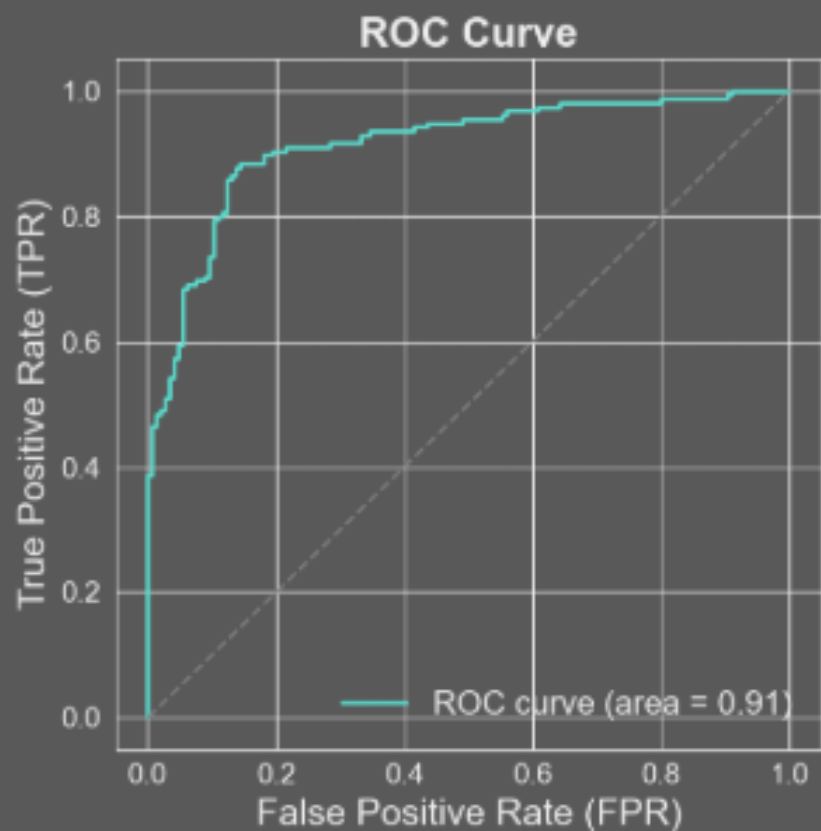
Model performance

Evaluation, decision boundaries, tuning, model comparisons

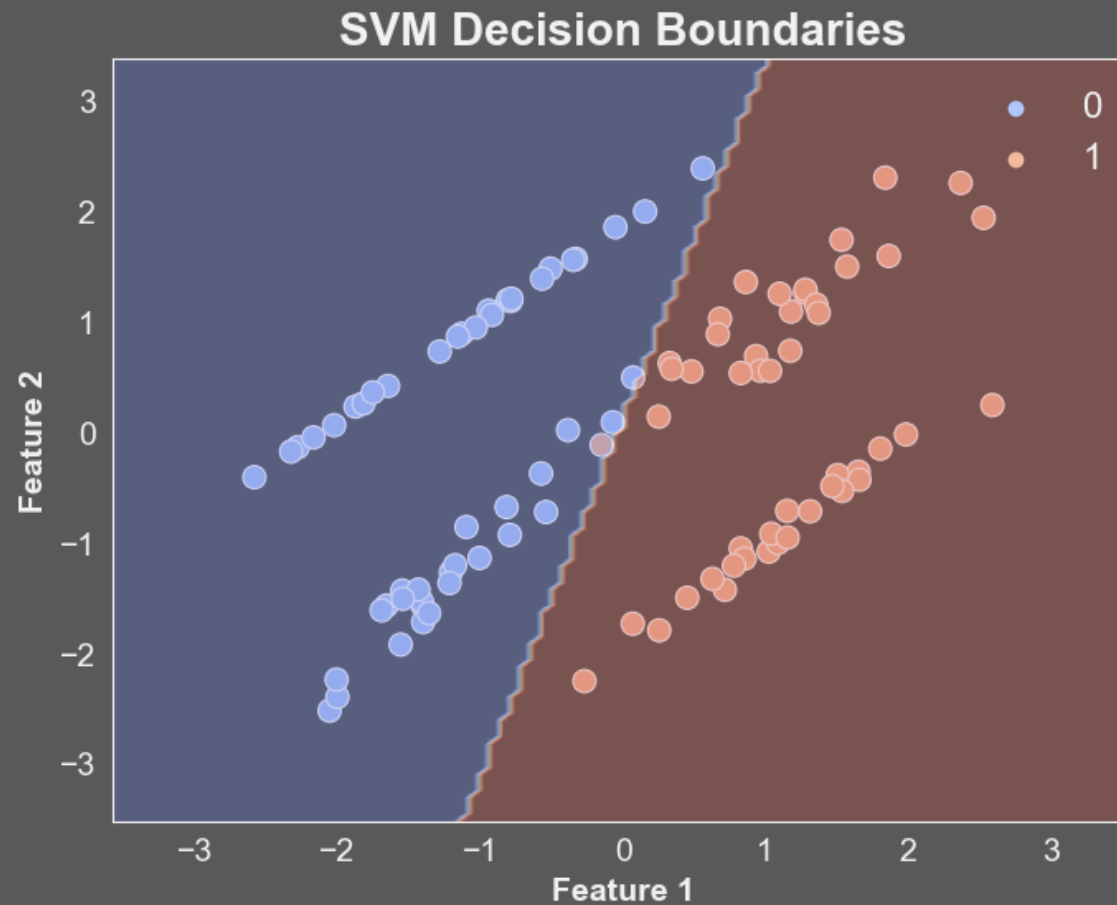
Regression: residual & histogram plots



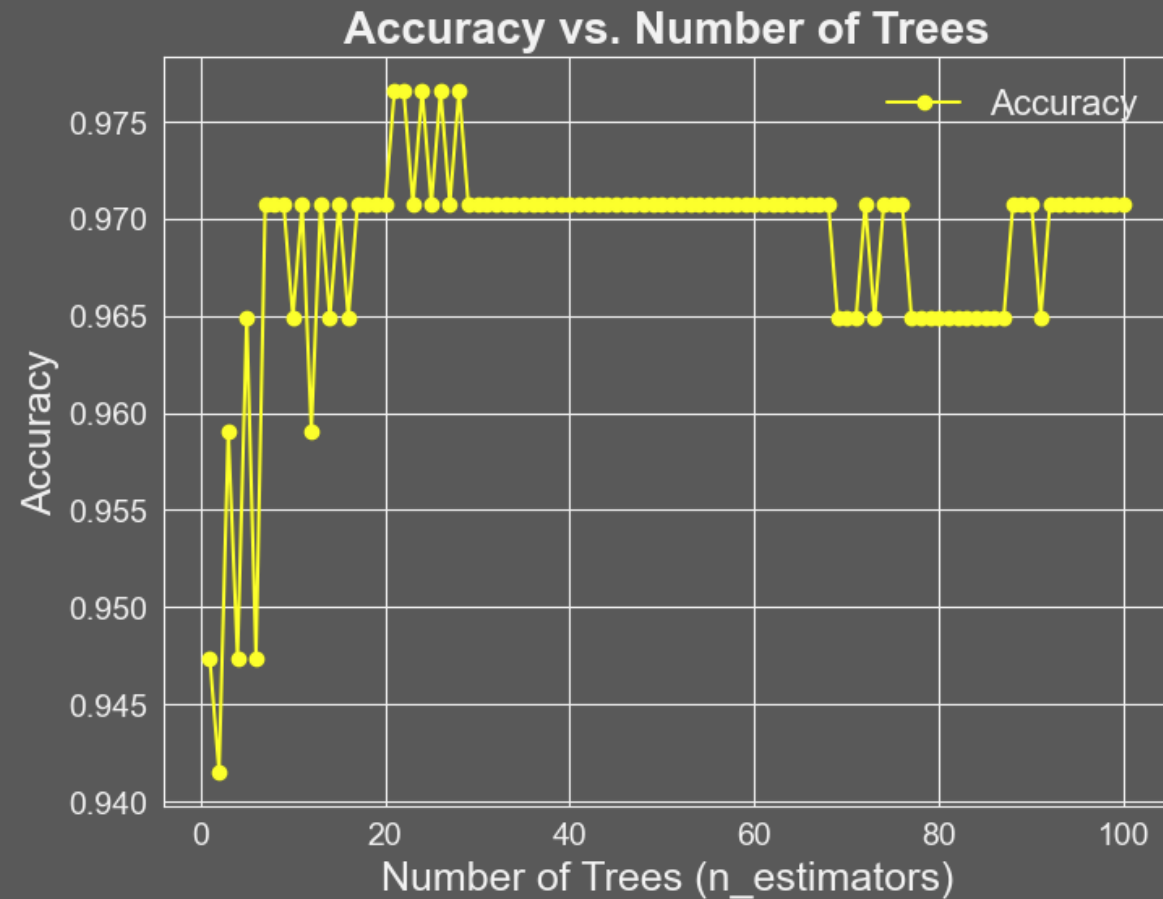
Classification: ROC and Confusion Matrix



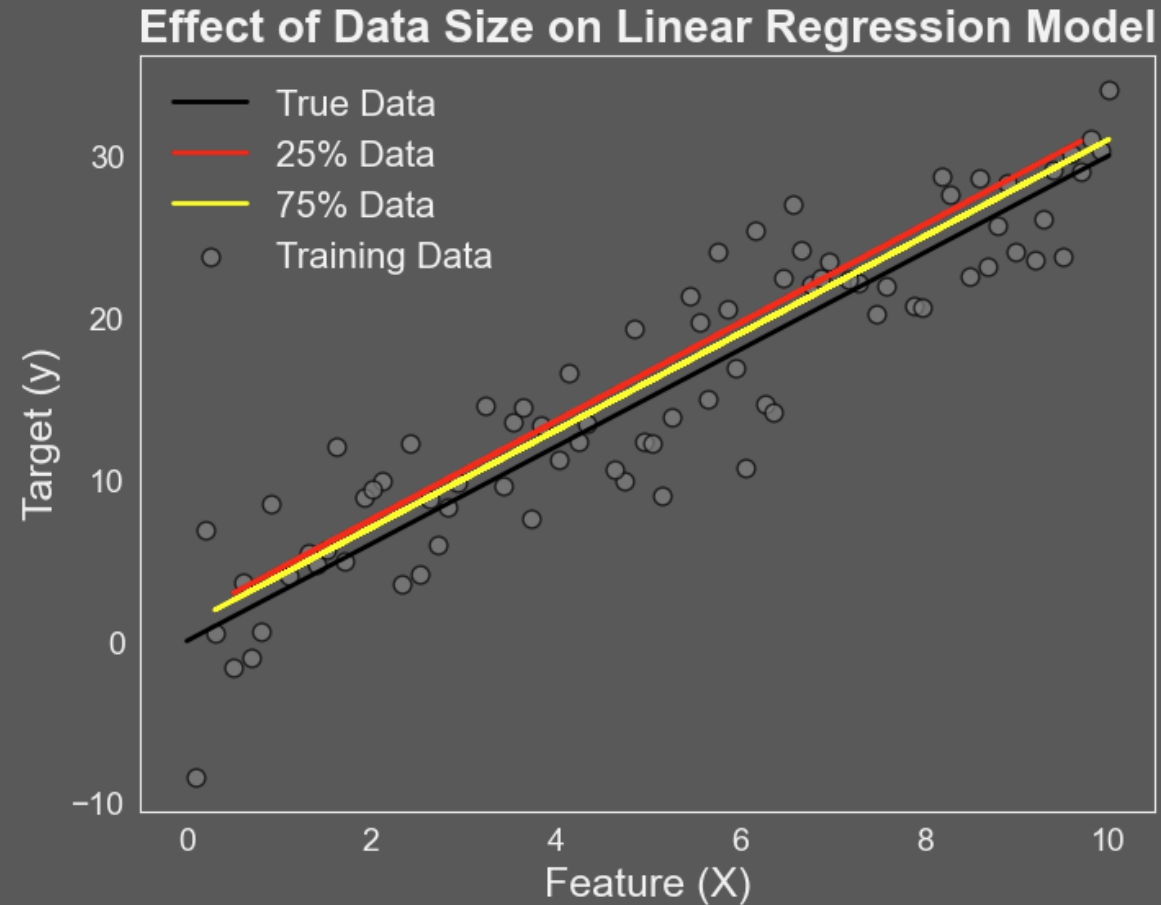
Decision boundaries



Model parameter tuning



Overfitting and underfitting



Assignments

- **Formative:**

Visualisation for machine learning using real-world datasets.

- **Summative:**

Explorable visual data stories (AE2)

Questions?

IP&V Course satisfaction survey 2025/26

<https://www.surveymonkey.com/r/75MVCYB>

